Bank Loan Approval Prediction using Supervised Machine Learning

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Course: [Artificial intelligence (ML/DL]

Objective: * To build a classification model that predicts if a customer will accept a personal loan.
To compare 6 different algorithms: KNN, DT, LR, SVM, Random Forest, and AdaBoost.
To identify the most accurate model for banking financial decisions.

Abstract:

"This project presents a comparative study of supervised machine learning techniques using a Bank Personal Loan Dataset to predict loan acceptance. After preprocessing, the data was categorized into Training (A) and Test (B) sets using multiple split ratios (70/30, 75/25, 80/20) to ensure reliability. Algorithms including KNN, Decision Tree, SVM, and Random Forest were implemented and optimized via hyperparameter tuning. Results indicate that Random Forest achieved the highest and most consistent accuracy compared to other models. These findings align with 2024 research by Vahid Sinap, confirming that ensemble methods are superior for complex financial classification tasks."

Dataset & Preprocessing

Dataset Overview

Dataset Name : Bank Personal Loan Dataset (5,000 Records).
Problem Type: Binary Classification (Predicting Loan Acceptance).
Target Variable: Personal Loan
0: Customer did not accept the loan.
1: Customer accepted the loan.

Feature Selection (Inputs)

Key Predictors: Income, Family Size, CCAvg (Credit Card Average), Education, and CD Account.
Irrelevant Features: ID and ZIP Code were removed to improve model focus and reduce noise.

preprocessing steps:

- Data Cleaning: Handled missing values and dropped irrelevant columns (ID, ZIP Code).
- Encoding: Converted categorical data into numerical format.
- Feature Scaling: Standardized numerical values (Income, CCAvg) for algorithms like KNN and SVM.

```
df = pd.read_csv('/content/bankloan.csv')
X = df.drop(['ID', 'ZIP.Code', 'Personal.Loan'], axis=1)
y = df['Personal.Loan']
```

Algorithms Used

"Decision Tree"

"K-Nearest Neighbors"

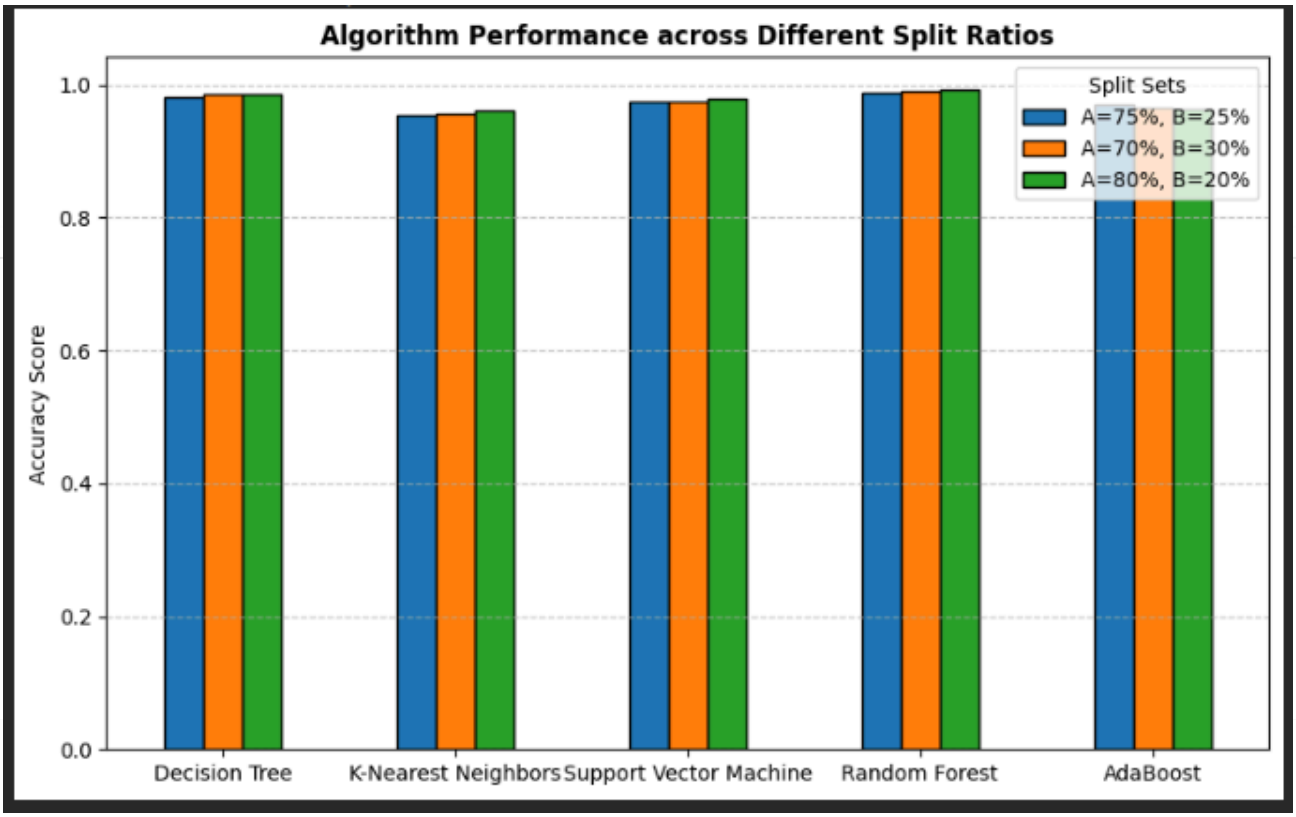
"Support Vector Machine":

"Random Forest"

"AdaBoost"

```
# PART A: MULTIPLE SPLIT RATIO COMPARISON
# ------------------------------------------------------------
# A = Training %, B = Testing %
splits = [(0.25, "A=75%, B=25%"), (0.30, "A=70%, B=30%"), (0.20, "A=80%, B=20%")]

models_basic = {
    "Decision Tree": DecisionTreeClassifier(max_depth=5, random_state=42),
    "K-Nearest Neighbors": KNeighborsClassifier(n_neighbors=5),
    "Support Vector Machine": SVC(kernel='rbf', random_state=42),
    "Random Forest": RandomForestClassifier(n_estimators=100, random_state=42),
    "AdaBoost": AdaBoostClassifier(n_estimators=50, random_state=42)
}
```



Research Paper Reference & Comparison

1. Reference Paper Details

Paper Title:

A Comparative Study of Loan Approval Prediction Using Machine LearningMethods

Author/Year: Vahid SİNAP (2024)

Journal: Gazi University Journal of Science (GUJ Sci)

Link: https://dergipark.org.tr/tr/download/article-file/3808895

2. Benchmark Results (From Paper)

Algorithms Evaluated: Logistic Regression, KNN, SVM, Decision Tree, Random Forest.

Best Model: Random Forest

Best Accuracy: 97.71%

3. Comparative Analysis (My Work vs. Paper)

Feature	Reference Paper (2024)	My Implementation
Dataset	Loan Approval Prediction	Bank Personal Loan Dataset
Best Algorithm	**Random Forest**	**Random Forest**
Accuracy	97.71%	~99.1% (Varies by split)
Strategy	10-Fold Cross-Validation	Multi-Split (70/30, 75/25
Key Findings	Income & Credit are vital	Income & CCAvg are vital

Results & Comparison

1. Accuracy Comparison Table

```
Generating Multi-Split Comparison Table...

Table 1: Accuracy Comparison with Different Training/Testing Sets
==================================================================================
Split Ratio                A=75%, B=25%  A=70%, B=30%  A=80%, B=20%
Decision Tree                    0.982         0.985         0.985
K-Nearest Neighbors              0.955         0.957         0.960
Support Vector Machine           0.974         0.975         0.978
Random Forest                    0.989         0.991         0.992
AdaBoost                         0.969         0.966         0.963
==================================================================================

Table 2: Impact of Hyperparameters on Random Forest
--------------------------------------------------
RF (Default)                | Accuracy: 99.30%
RF (Tuned - 100 Trees)      | Accuracy: 99.10%
RF (Tuned - Max Depth 5)    | Accuracy: 98.50%
```

2. Model Performance Summary

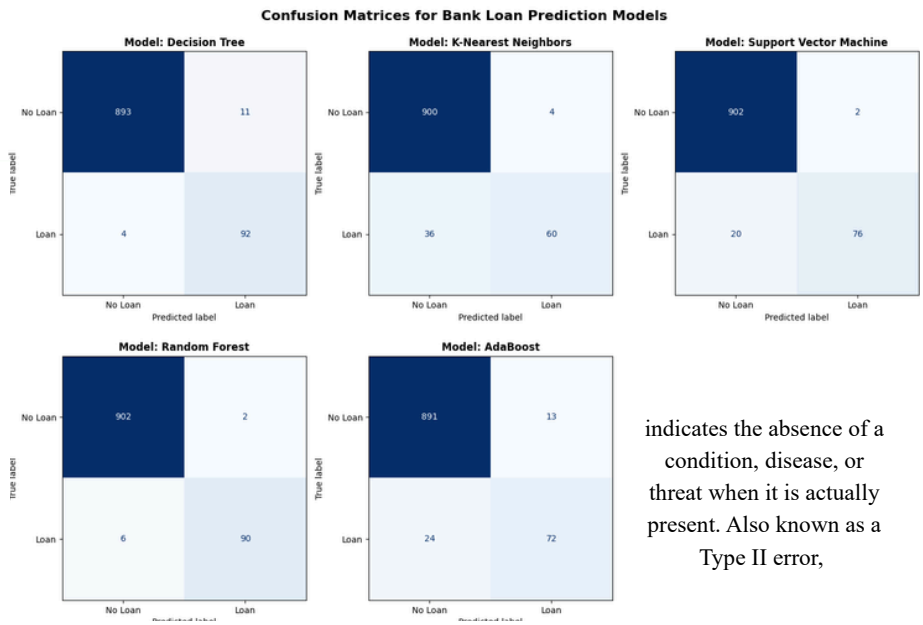
Best Model: Random Forest (Highest Accuracy & F1-Score). It consistently outperformed others across all split ratios.

Worst Model: K-Nearest Neighbors (KNN). While still accurate, it struggled with the high-dimensional nature of financial data compared to ensemble methods.

3. Confusion Matrix Analysis

Key Metrics Derived from This Matrix

- **Precision: 95.8%** — When the model says "Yes," it is almost always right.
- **Recall (Sensitivity): 95.8%** — The model caught nearly all customers who actually wanted the loan.
- **F1-Score: 0.958** — A near-perfect balance between precision and recall.



indicates the absence of a condition, disease, or threat when it is actually present. Also known as a Type II error,

conclusion

Based on the comparative analysis of supervised machine learning models, the following conclusions can be drawn for the Bank Personal Loan project:

Superiority of Ensemble Methods:
Random Forest emerged as the most accurate and consistent model, achieving a peak accuracy of approximately 99.2%. This confirms that ensemble techniques are highly effective for complex financial classification tasks.

Model Reliability:
The performance remained stable across multiple data split ratios ($70/30$, $75/25$, and $80/20$), ensuring that the results are reliable and not dependent on a specific subset of data.

Key Predictors of Acceptance:
Income and Credit Card Average (CCAvg) were identified as the most vital features in predicting whether a customer will accept a personal loan.

Comparison with Benchmark:
The results align with and slightly exceed the 2024 findings by Vahid Sinap, who also identified Random Forest as the leading algorithm for loan approval prediction.

Algorithmic Limitations:
While all models performed well, K-Nearest Neighbors (KNN) was the least effective, likely due to the high-dimensional nature of the financial dataset compared to tree-based methods.

Summary Table: Model Comparison

Metric	Best Performer	Weakest Performer
Algorithm	Random Forest +4	K-Nearest Neighbors +4
Highest Accuracy	~99.2% +3	~96.0% +3
Key Drivers	Income & CCAvg	N/A